First control of Computer Networks (XC), Bachelor in Computer Engineering (GEI)		07/04/2025	Spring 2024-2025
NAME:	**SURNAME(S):**	**GROUP**	**DNI**

Duration: 1h30m. The test will be completed in 25 minutes. Answer the problems in the same sentence.

Test. (3 points) Questions are worth half if there is one mistake and 0 if there is more than one.

1. Mark the correct statements about IP networks:
- ☐ The network address 130.0.0.0/8 is of Class A.
- ☐ The network address 192.168.50.0/24 is private.
- ☐ The subnetwork address 172.16.32.0/20 is valid.
- ☐ The network address 0.0.0.0/0 in the routing table represents the devices on a network.

2. Mark the correct statements about the 192.168.64.128/26 network:
- ☐ The subnet mask is 255.255.255.192.
- ☐ The smallest unicast address is 192.168.64.129.
- ☐ The network's broadcast address is 192.168.64.255.
- ☐ The network's broadcast address is 192.168.64.191.

3. About the IP protocol:
- ☐ It is a Layer 2 protocol of the OSI model.
- ☐ It guarantees that all packets will be received by the destination.
- ☐ When an IPv4 packet is fragmented, upon reaching the destination, all fragments of the packet have the same fragment identifier.
- ☐ IPv4 addresses always have 16 bits.

4. About the ICMP (Internet Control Message Protocol) protocol:
- ☐ The ping and traceroute commands use ICMP.
- ☐ ICMP packets do not pass through a router with PNAT.
- ☐ ICMP packets are sent in the payload of the transport protocol.
- ☐ It provides error messages.

5. About the ARP (Address Resolution Protocol):
- ☐ Clients start by sending broadcast messages to the MAC address 255.255.255.255.
- ☐ It is used to find the MAC address associated with an IP address on the local network.
- ☐ The ARP Request message is unicast.
- ☐ “Gratuitous ARP” is used to detect duplicate IP addresses.

6. About the DHCP (Dynamic Host Configuration Protocol) protocol:
- ☐ It configures various client parameters, not just the IP address.
- ☐ The server sends DHCPDISCOVER and DHCPREQUEST messages to the broadcast address 255.255.255.255.
- ☐ Clients must know the server's IP address (unicast).
- ☐ It uses port 80 to connect to the server.

7. In a routing table:
- ☐ Each entry contains the destination address, mask, gateway, and interface.
- ☐ It can contain both static and dynamic entries.
- ☐ The values in the “Gateway” column can be empty or contain the IP address of the next router through which the packet is to be sent.
- ☐ If the table doesn't have an entry for a destination, the packet is discarded and an ICMP message of the type “network unreachable” is sent.

8. About RIP (Routing Information Protocol) version 2:
- ☐ RIP Update messages include the destination and metric and are sent every 30 seconds.
- ☐ RIP Update messages are sent to the broadcast address.
- ☐ The "Split horizon" method is used to reduce the effect of "Count to infinity."
- ☐ The infinite metric (Destination unreachable) is 16.

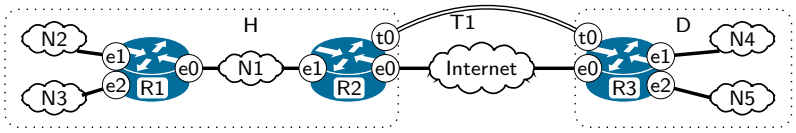
9. About NAT (Network Address Translation):
- ☐ When using Dynamic NAT, entries in the NAT table do not expire.
- ☐ When PAT (Port and Address Translation), also called PNAT, acts on packets that go out to the Internet, it changes the source IP while always maintaining the destination port.
- ☐ When PAT (Port and Address Translation), also called PNAT, acts on packets that go out to the Internet, it changes the source IP and can change the source port.
- ☐ DNAT (Destination Network Address Translation) allows a device on a private network to connect to another device on the public network.

10. About the client-server communication model:
- ☐ The value of an ephemeral port is less than 1024.
- ☐ Packets in a communication between client and server processes are identified by source and destination IP addresses, client and server ports, and protocol.
- ☐ A device can only establish one client connection to the same server and protocol.
- ☐ A device can act as both, a client and a server.

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Duration: 1h30m. All questions have a single true answer. Mark the answers on the back.

Problem 1 7 points.



N1: 192.168.0.0/28 ISP-H: 80.0.0.1
N2: 192.168.0.128/27 R2-e0: 80.0.0.2
N3: 192.168.0.192/26 ISP-D: 90.0.0.1
T1: 192.168.1.0/24 R3-e0: 90.0.0.2

In the network in the figure there is a headquarters H and a branch office D connected by an IPIP tunnel (T1). To access the Internet, NAT is configured and used only when necessary.

To configure networks N4 and N5, we want to subnet the network 192.168.2.0/24 so that 50 and 80 PCs can be placed on N4 and N5, respectively. We want the numerically lower addresses and that the maximum number of consecutive free IPs remains (unassigned to N4,N5). State what the subnets will be (IP/masc). Justify your answer.

1. N4: A. 192.168.2.0/27 B. 192.168.2.0/25 C. 192.168.2.0/26 D. 192.168.2.0/24
2. N5: A. 192.168.2.64/25 B. 192.168.2.128/26 C. 192.168.2.64/26 D. 192.168.2.128/25

Suppose the numerically smallest IPs are assigned to the router interfaces in the order R1, R2, R3. State what the IPs assigned to the router interfaces will be:

3. R1.e1: A. 192.168.0.128 B. 192.168.0.1 C. 192.168.0.129 D. 192.168.0.130
4. R2.e1: A. 192.168.1.1 B. 192.168.0.1 C. 192.168.0.0 D. 192.168.0.2
5. R3.t0: A. 192.168.1.2 B. 192.168.1.0 C. 192.168.1.1 D. 192.168.0.1

Say what the broadcast addresses of the following networks are.

6. N2: A. 192.168.0.159 B. 192.168.0.63 C. 192.168.0.255 D. 192.168.0.151
7. N4: A. 192.168.2.255 B. 192.168.2.31 C. 192.168.2.127 D. 192.168.2.63

Routing tables are configured manually (static routing). State what entries will be in the routers' routing tables if the networks are aggregated using the following criteria (and in this order from highest to lowest preference): 1) there must be connectivity between networks N1,···,N5 and access to the Internet; 2) there must be the minimum number of entries (tables as small as possible); 3) the masks must be as large as possible (e.g. /25 is preferred to /24). Notation: (destination/mask; gateway; interface), B=192.168, and N1,N2,N3,T1 are the networks in the figure.

8. R1: A. (0.0.0.0/0; R2.e1; e0) B. (T1; R2.e1; e0) C. (B.0.0/16; R2.e1; e0) D. (B.2.0/24; R2.e1; e0)
9. R1: A. (N1; R2.e1; e0) B. (N4; R2.e1; e0) C. (80.0.0.0/24; R2.e1; e0) D. (N1; -; e0)
10. R2: A. (0.0.0.0/0; 80.0.0.1; t0) B. (0.0.0.0/0; R3.t0; t0) C. (0.0.0.0/0; 80.0.0.1; e0) D. (0.0.0.0/0; R3.e0; e0)
11. R2: A. (B.0.0/24; R1.e0; e1) B. (B.0.128/25; R1.e0; e1) C. (B.0.0/26; R1.e0; e1) D. (N1; R1.e0; e1)
12. R2: A. (B.2.0/25; R3.t0; t0) B. (B.2.0/24; R3.t0; e0) C. (B.2.128/25; R3.t0; t0) D. (B.2.0/24; R3.t0; t0)
13. R3: A. (N1; R2.t0; t0) B. (B.0.128/25; R2.t0; t0) C. (B.0.0/24; R2.t0; e0) D. (B.0.0/24; R2.t0; t0)
14. R3: A. (T1; 90.0.0.1; t0) B. (T1; -; t0) C. (T1; R2.t0; t0) D. (T1; 90.0.0.1; e0)

We want to expand H by adding another subnet to R1 to which the maximum number of PCs (max PCs) can be connected by taking the IPs of 192.168.0.0/24 that are free (not assigned to other networks). Give the following information about the new subnet.

15. network address/mask: A. 192.168.0.0/24 B. 192.168.0.128/25 C. 192.168.0.64/26 D. 192.168.0.192/26
16. broadcast address: A. 192.168.0.63 B. 192.168.0.191 C. 192.168.0.255 D. 192.168.0.127
17. max PCs: A. 61 B. 125 C. 64 D. 29

PCs A and B have the largest IP that can be assigned in N2 and N4, respectively. A pings B. Say what the IPs and protocol of the IP header of the echo request will be when it reaches R3 on e0.

18. external header, source address: A. 80.0.0.2 B. 192.168.0.158 C. 192.168.2.62 D. 90.0.0.2
19. external header, destination address: A. 90.0.0.2 B. 80.0.0.2 C. 192.168.2.62 D. 192.168.0.158
20. external header, protocol: A. ICMP B. UDP C. TCP D. IPIP
21. internal header, source address: A. 90.0.0.2 B. 80.0.0.2 C. 192.168.0.158 D. 192.168.2.62
22. internal header, destination address: A. 192.168.2.62 B. 80.0.0.2 C. 192.168.0.158 D. 90.0.0.2
23. internal header, protocol: A. TCP B. UDP C. IPIP D. ICMP

Suppose that in the previous question B is off. Say whether it is plausible that any datagram will reach A due to the ping. If so, say who will send it and the type of message the datagram will carry.

24. A. Yes, R2 B. Yes, R3 C. Yes, R1 D. No
25. A. an ICMP echo reply message B. an ICMP error message C. an UDP error message D. no message

Now PC A sends a DNS request to the Internet name server 8.8.8.8. What will be the following values of the IP header when the datagram arrives at R2 on e1 (R2.e1-in) and leaves R2 on e0 (R2.e0-out)?.

26. R2.e1-in, source address: A. 8.8.8.8 B. 192.168.1.1 C. 192.168.0.158 D. 80.0.0.2
27. R2.e1-in, destination address: A. 192.168.1.1 B. 80.0.0.2 C. 8.8.8.8 D. 192.168.0.158
28. R2.e0-out, source address: A. 80.0.0.2 B. 8.8.8.8 C. 192.168.0.158 D. 192.168.1.1
29. R2.e0-out, destination address: A. 192.168.0.158 B. 8.8.8.8 C. 192.168.1.1 D. 80.0.0.2
30. internal header, protocol: A. ICMP B. UDP C. TCP D. IPIP

On a PC there is the ARP table that shows the dump. Say if with this information we can know which network the PC is connected to. If so, say which network it is.

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$ arp -n
Address          HWaddress           Iface
192.168.0.197    00:0d:b9:4d:12:30   eth0
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31. A. Yes, N3 B. Yes, N2 C. Yes, N1 D. No

Suppose that the previous PC then executes the command ping 192.168.0.1. Once the PC receives the ping response, say whether any IP address will have been added to the PC's ARP table, and if so, which one.

32. A. No B. Yes, 192.168.0.193 C. Yes, 192.168.0.192 D. Yes, 192.168.1.65